3 **(a)** Define *electric potential* at a point.

.....

.....

..... [1]

(b) Two similar positive point charges of magnitude *e* are set up along the line XY as shown in Fig. 3.1.

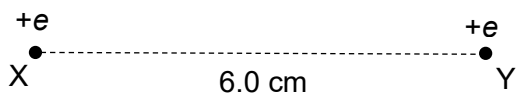

Fig. 3.1

(i) On Fig 3.2, sketch how the electric potential varies over XY. [2]

Fig. 3.2

(ii) Describe how the electric force acting on a known charge *q* placed at a point along XY could be obtained from Fig 3.2.

.....

..... [2]

(c) **(i)** A negative point charge of – *e* is now placed at the midpoint of XY.

State the magnitude of electric force experienced by the point charge

(ii) The negative point charge at the midpoint in XY in **(c)(i)** is moved to the position Z as shown in Fig 3.3.

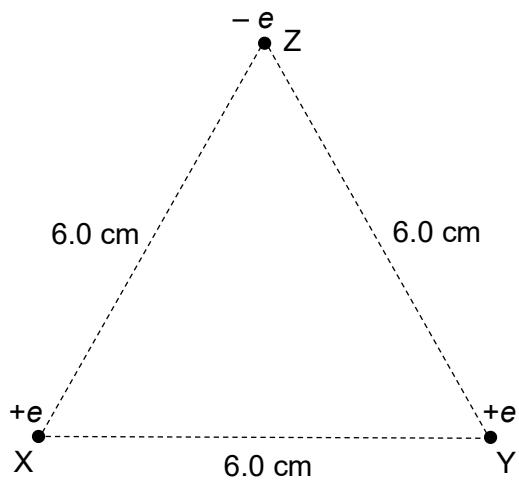

Fig. 3.3

Within the bounded region of Fig 3.4, sketch the resultant electric field lines of the 3 point charges. [3]

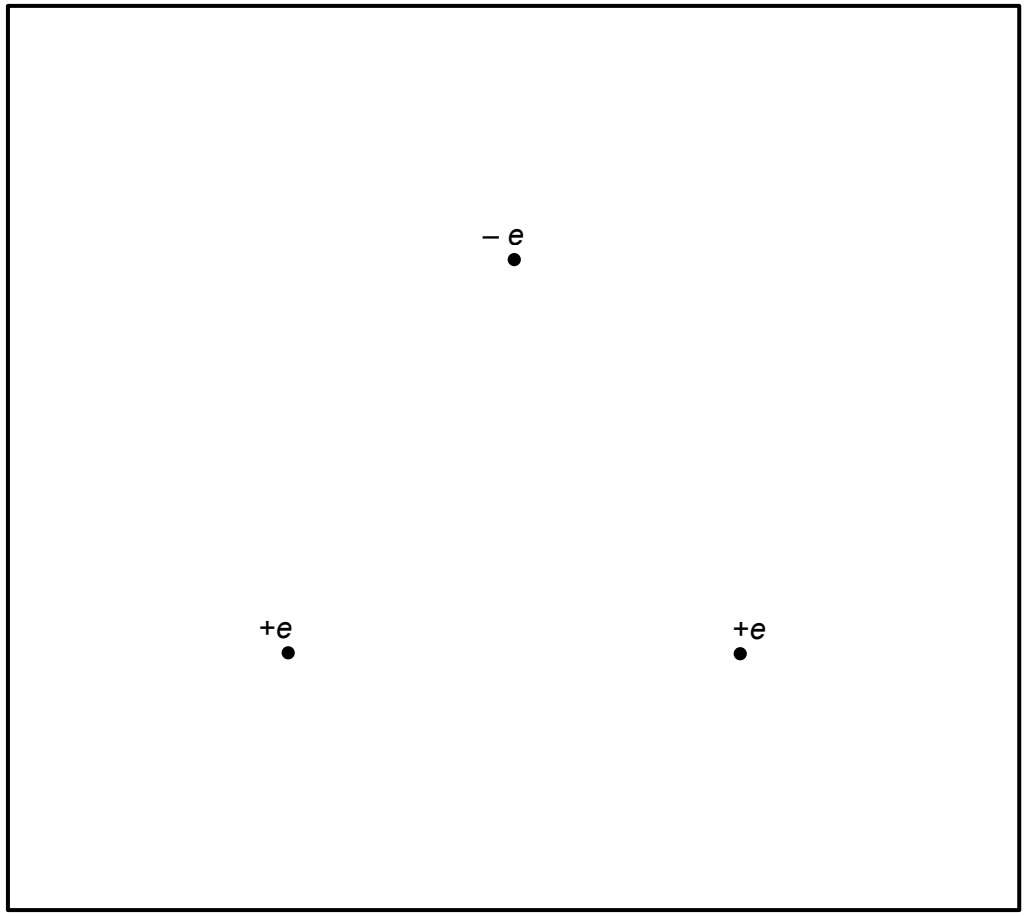

Fig. 3.4

(iii) Determine the work done by the electric field to completely separate the 3 point charges in Fig 3.3.

work done = ... J [3]